

Government Procurement Law and Firm Innovation^{*}

Yiqing Yuan[†]

November 2025

Abstract

Over the past decades, the scale of government procurement has been steadily expanding, playing an important role in resource allocation and technological progress in China. Using the implementation of the 2003 Chinese Government Procurement Law (GPL) as a quasi-natural experiment, I employ a difference-in-differences design to examine how tighter regulation of government procurement affects firms' technological innovation. The results indicate that the GPL significantly increased firms' patent applications and R&D investment in procurement-intensive industries. The effect is stronger in cities with higher pre-existing corruption, and among firms with prior R&D expenditures, previous patenting, or higher total factor productivity, which are typically more efficient firms. This pattern is consistent with the GPL's objective to promote an open procurement market and reallocate government contracts toward more efficient firms, particularly in places that were previously more corrupt and distorted. By doing so, the GPL has strengthened firms' incentives to invest in R&D, and fostered innovation growth. I further propose a research design to investigate the underlying mechanisms more granularly in a more recent setting, using a novel contract-level dataset.

Keywords: government procurement; technological innovation; resource misallocation; Government Procurement Law of the People's Republic of China

^{*}This work was supported by Junzheng Fund, Fudan University Undergraduate Research Opportunity Program (Grant number: 24947).

[†]School of Economics, Fudan University.

1 Introduction

Governments often act as the largest buyer in the economy through government procurement. Around the world, government procurement amounts to about 12% of global GDP in 2018 (Bosio et al., 2022). In China, its scale has also been steadily expanding, reaching 3.70 trillion yuan (0.52 trillion US dollars) in 2020, which accounts for approximately 10.2% of China’s fiscal spending (Figure 1). Given its scale, government procurement greatly affects resource allocation, playing an important role in firm growth and technological progress. However, government procurement is especially vulnerable to corruption, with benefit transfers pervasive in a nontransparent market (OECD, 2016). Previous studies have shown that corruption has led to resource misallocation in industries that rely more on government procurement, at the cost of both public funds and social welfare (Hang and Zhan, 2023). However, firm-level studies remain limited, especially regarding the impact of the misallocation of procurement contracts on firms’ technological innovation.

This paper utilizes the implementation of the 2003 Chinese Government Procurement Law (the GPL, hereafter) as a quasi-natural experiment to investigate the causal relationship between the misallocation of procurement contracts and firms’ innovation. Before the GPL was implemented, government procurement in China was under-regulated, with procurement contracts assigned in nontransparent ways. The GPL was the first nationwide law to mandate open auctions as the principal procurement method, requiring contracts to be allocated based on product quality and price, following standardized procedures in a transparent market. The GPL has successfully reduced the importance of non-market factors in the allocation of procurement contracts and reduced resource misallocation, as documented by Hang and Zhan (2023).

Building on this institutional change, I use a difference-in-differences design to investigate how the GPL affects firms’ technological innovation. To do this, I combine three main databases: (i) the China Investment Climate Survey (CICS), which is used to impute the procurement intensity of each industry as the treatment variable; (ii) the Above-Scale Industrial Firms (ASIF) of China (1998–2013), which provides firm-level economic data covering both pre- and post-GPL years; and (iii) China’s Patent Data (1998–2013), which supplement the ASIF data with detailed patent application records used to measure firms’ technological innovation.

I find a positive and persistent effect of government procurement regulation on firms’ innovation growth. After the implementation of the GPL, firms’ innovation lev-

els in procurement-intensive industries significantly increased. The effect is greater for firms with prior R&D expenditures, prior patent application experience, or higher total factor productivity, which are typically more efficient firms. I further observe that firms’ R&D investment in procurement-intensive industries increased, and that the effect is stronger in more corrupt cities. This pattern is consistent with the objective of the GPL, which was to promote an open procurement market and reallocate government contracts toward more efficient firms, particularly in places that were previously more corrupt and distorted. By doing so, the GPL has strengthened firms’ incentives to invest in R&D, bringing about innovation growth.

Due to data limitations, this paper cannot provide direct evidence on firm-level contract allocation. To address this, I outline a future research design that allows for more granular analysis, as discussed in detail in Section 7. Specifically, I have constructed a novel contract-level dataset, which records all government procurement contracts in the province and contains rich information on procuring entities, participating firms, bid scores, and bid values. This dataset will enable accurate identification of corrupt contracts and the tracing of firms’ subsequent performance. I also use the data to examine how winning government procurement contracts affects firms’ subsequent innovation, in a quasi-random “close auction” setting.

The remainder of the paper is organized as follows. Section 2 reviews related literature, Section 3 presents the institutional background of the GPL, Section 4 describes the empirical strategy, and Section 5 reports the results. Section 6 concludes, and Section 7 outlines a future research design.

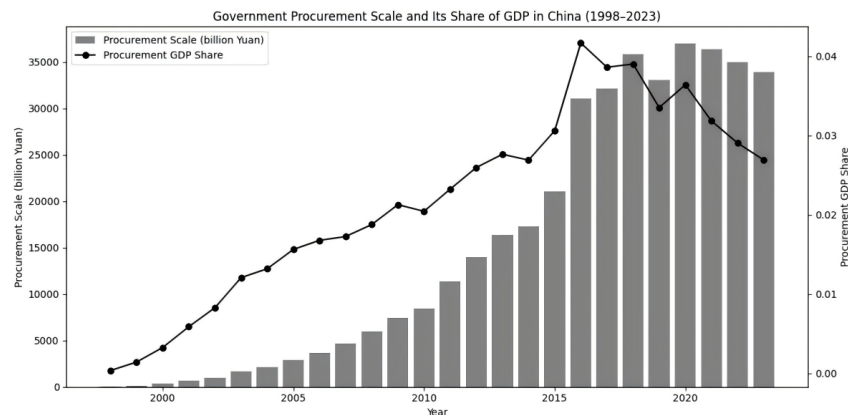


Figure 1. Government Procurement Scale and Share of GDP, 1998–2023

Notes: The data on government procurement amounts are from the China’s Ministry of Finance; GDP data are from China’s National Bureau of Statistics. Both series are reported in nominal RMB.

2 Literature Review

This paper mainly relates to three strands of literature: corruption in government procurement contracts, resource misallocation and firm innovation, and policies and innovation in China.

2.1 Corruption in Government Procurement Contracts

The literature on government procurement shows that corruption is rampant in the procurement process and that contract allocation is often distorted. Evidence shows that politically connected firms may be more likely to win orders, depressing the performance of others (Coviello and Gagliarducci, 2017). Firms may also use political donations to obtain contracts, crowding out other firms' resources (Baltrunaite, 2020; Titl and Geys, 2018). This distortion may further lead to the misallocation of production factors and lower efficiency, since firms with stronger political connections tend to be less productive. Brugués et al. (2022) study the impact of firms' political connections on resource allocation in Ecuador and evaluate the welfare cost of misallocated procurement orders. Hang and Zhan (2023) study government procurement regulation in China and find that it reduces contract misallocation, providing the empirical background for this paper.

Other studies find that in non-democratic contexts, distortions may also arise from the political risks of government officials. Li and Liu (2024) document a risk-driven favoritism toward state-owned enterprises in government procurement during China's anti-corruption campaign, as officials aim to signal conformity with regime insiders, which further leads to allocation distortions and lower firm productivity.

2.2 Resource Misallocation and Firm Innovation

A rich body of work shows that reallocating resources to more efficient and innovative firms can drive productivity growth (Jovanovic, 1982; Hopenhayn, 1992; Ericson and Pakes, 1995). However, policy frictions can impede this allocation efficiency, lower productivity, and weaken firms' incentives to innovate. For example, Hopenhayn and Rogerson (1993) find that firing costs prevent unproductive firms from shrinking, leading to lower productivity growth. Acemoglu et al. (2018) find that subsidies to the R&D of incumbents can inefficiently encourage the survival and expansion of low productivity firms, whereas taxing the continued operation of incumbents can lead to sizable gains by encouraging the exit of less productive firms and freeing up skilled

labor for the R&D of high-productivity firms. [König et al. \(2022\)](#) find that pervasive output wedges in China often induce the wrong firms (and deter the right ones) from investing in R&D, reducing the overall returns to innovation.

This paper contributes by providing empirical evidence that government procurement regulation can reduce contract misallocation, allocate contracts to more efficient firms, and encourage R&D investment.

2.3 Innovation Policies in China

Finally, this paper relates to a growing literature on innovation policies in China. Existing work typically groups innovation policies into input-side policies that lower the cost of innovation and output-side policies that raise the returns to innovation. On the input side, government R&D subsidies relax firms’ financing constraints and strengthen both R&D investment and patenting performance in China ([Li, 2012](#); [Kou and Liu, 2020](#)). Bank competition induced by the relaxation of financial market regulations likewise eases financing constraints and promotes technological innovation among Chinese enterprises ([Chen and Zhang, 2019](#)). Also, preferential corporate tax policies have been found to lower the user cost of R&D and significantly increase patenting, especially for larger firms and those in more developed regions ([Shao and Xiao, 2019](#)). The “High-Tech Enterprise” certification, which provides tax breaks and improves access to external finance and market attention, further reshapes firms’ innovation incentives ([Chen et al., 2021](#)). On the output side, stronger intellectual property rights protection enhances firms’ ability to appropriate returns from innovation, inducing higher patenting and better innovation performance, particularly for privatized and private firms ([Fang et al., 2017](#); [Wu and Tang, 2016](#)).

Building on this literature, this paper offers a complementary perspective by emphasizing an indirect policy channel through resource allocation, providing new evidence and meaningful policy implications for the design of innovation institutions.

3 Background

Government procurement refers to the purchase of goods, construction projects, and services by all levels of government and their affiliated institutions using fiscal funds. In the 1990s, the legal environment of government procurement was underdeveloped in China: corruption was rampant, and the use of public funds was inefficient. For example, firms with political connections could sell large quantities of low-quality,

overpriced goods to the government, while government agencies received sizable kick-backs in this process.¹

To standardize procurement and reduce inefficient uses of fiscal funds, China implemented the Government Procurement Law of the People’s Republic of China on January 1, 2003, the first nationwide legislation governing the Chinese procurement market.² The GPL aims at standardizing government procurement practices and was part of a broader set of reforms to promote a market-oriented economic system in China. It was also viewed as a step toward meeting the requirements of the WTO Agreement on Government Procurement.

Specifically, the GPL mandates that open auctions should be the principal method of government procurement and formalizes procurement procedures. In open auctions, firms submit confidential bids, and contracts are awarded based on the highest combined scores of product quality and price. To promote fair competition, a minimum number of bidders, commonly no fewer than three, is mandated, and experts are randomly selected from an approved pool, with identities unrevealed until the bid opening. The GPL also establishes a comprehensive supervision and liability system: suppliers may file inquiries about procurement and administrative appeals with the supervising agency or the courts; legal liabilities for collusion and other violations are specified. These standardization efforts make all firms compete in an open and fair procurement market, largely reducing the misallocation of procurement contracts.

The GPL was successfully implemented, with substantial savings of fiscal funds.³ As the GPL comprehensively specifies procurement procedures, methods, and legal liabilities of all parties, it plays an important role in promoting a more open and transparent market and reducing contract misallocation. This paper utilizes the implementation of the GPL as a quasi-natural experiment and investigates its impact on firms’ technological innovation.

¹See <https://news.sina.com.cn/c/2001-10-29/387570.html>.

²The Tendering and Bidding Law was enacted in 2000, but only regulates large-scale public works projects and procurement by state-owned enterprises (SOEs). By contrast, the GPL focuses on procurement of projects, services, and goods by government (different from SOE procurement in China, which refers to state-owned enterprises’ purchases for their production, investment, or operation).

³At the 2005 National Government Procurement Work Conference, it was noted that, following implementation, specialized institutes for province-level government procurement had expanded and saved substantial fiscal resources. See https://www.ccg.gov.cn/ldjh/201711/t20171108_9124428.htm.

4 Empirical Design

4.1 Identification Strategy

To identify the causal relationship, I adopt a difference-in-differences (DID) strategy, using the implementation of the GPL as a quasi-natural experiment. The treatment group consists of industries with government procurement intensities higher than the median. The baseline specification is:

$$Y_{ijct} = \beta (gs_j \times post03_t) + \mathbf{X}_i^{\text{pre}} \cdot \boldsymbol{\lambda}_t + \mathbf{Z}_{jt} \cdot \boldsymbol{\delta} + \mu_i + \phi_{ct} + \psi_{ot} + \varepsilon_{ijct}, \quad (1)$$

where Y_{ijct} represents firm i 's innovation outcome in year t , belonging to four-digit industry j and located in city c , measured by patent application numbers (both in natural logarithm form and in levels). The variable gs_j is an indicator for the treatment status, equal to one if industry j has above-median government procurement share and zero otherwise. The variable $post03_t$ is a dummy indicating the post-GPL period, equal to one from 2003 onward and zero otherwise.

The vector $\mathbf{X}_i^{\text{pre}}$ contains pre-policy firm-level characteristics (the natural logarithm of firm assets, capital-labor ratio, age, and an exporter indicator), interacted with year dummies $\boldsymbol{\lambda}_t$. The vector $\mathbf{Z}_{jt}$ contains controls for other major concurrent events, including the WTO accession and the grant of permanent normal trade relations (PNTR) to China by the United States Congress in October 2000. The term μ_i denotes firm fixed effects, capturing unobserved, time-invariant heterogeneity across firms; ϕ_{ct} denotes city-year fixed effects, capturing differences in policy implementation and other shocks across cities over time; and ψ_{ot} denotes ownership-year fixed effects, capturing unobserved heterogeneity specific to ownership types over time. The error term is ε_{ijct} . Standard errors are clustered at the four-digit industry level.

The coefficient β is the main parameter of interest. A positive β indicates higher innovation in procurement-intensive industries after the GPL, relative to industries with low procurement intensities.

The identification assumption for this strategy is the standard parallel trends assumption: in the absence of the GPL, the outcome variables would have followed the same time trend across industries with different procurement intensities. That is, firms in procurement-intensive industries should have had the same trend of innovation growth as their counterparts. To validate the parallel trends assumption, I also

estimate an event-study specification:

$$Y_{ijct} = \sum_{k \neq 2002} \beta_k (gs_j \times \mathbf{1}\{t = k\}) + \mathbf{X}_i^{\text{pre}} \cdot \boldsymbol{\lambda}_t + \mathbf{Z}_{jt} \cdot \boldsymbol{\delta} + \mu_i + \phi_{ct} + \psi_{ot} + \varepsilon_{ijct}, \quad (2)$$

where all variables are defined as in equation (1), and the year 2002 (the year prior to the implementation of the GPL) is the omitted baseline. The set of coefficients $\{\beta_k\}$ measures the relative relationship between industry-level government procurement intensities and firms' innovation in year k , compared with 2002. If the parallel trends assumption holds, the estimated β_k should be approximately zero throughout the pre-GPL period and should change only in the post-GPL period.

4.2 Data

Above-Scale Industrial Firms (ASIF) of China (1998–2013). This is the main dataset providing comprehensive panel data on individual firms, which allows for the construction of key control variables. The database is compiled by the National Bureau of Statistics of China (NBSC), covering approximately 150,000 to 400,000 firms annually, and includes all state-owned enterprises and non-state enterprises with annual sales above 5 million yuan.⁴ These firms account for roughly 90% of national industrial output and 70% of employment. I drop all non-manufacturing firms to focus on the manufacturing sector.

China's Patent Data (1998–2013). This database is used to measure firms' technological innovation by patent applications, and can be merged with the ASIF data. The database is sourced from the China National Intellectual Property Administration (CNIPA) and contains detailed patent information recorded since the establishment of China's patent office in 1985, including year, applicant name, and patent descriptions.

China Investment Climate Survey (CICS) Data. This dataset is part of the World Bank Enterprise Surveys, covering 12,400 firms located in 120 cities. The survey includes both basic firm information, such as industry and city, and financial information such as total sales and employment. Most firms report information in 2004, whereas some report data back to 2002. I use the CICS data to compute each

⁴This threshold was adjusted to 20 million yuan after 2011.

industry’s government procurement intensity, following [Hang and Zhan \(2023\)](#), which underlies the treatment classification.

4.3 Variable Construction

Dependent variable: firm innovation. I measure firm i ’s technological innovation in year t using patent application numbers. I use both dummies indicating whether any patents were applied for (the extensive margin) and the log of application counts (the intensive margin). There are three patent types: invention patents, utility model patents, and design patents.

Treatment status: government procurement intensity. Following [Hang and Zhan \(2023\)](#), I use the CICS database to construct government procurement intensities. I compute each industry’s weighted share of sales to the government by aggregating firms’ total sales and the share of sales to the government at the industry level. I then determine treatment status by whether this share is above or below the median. CICS has surveys in 2003 and 2005. The 2003 wave has only 2,400 observations and covers six manufacturing sectors, whereas the 2005 wave has 12,400 observations covering all two-digit manufacturing industries and all regions except Tibet. I therefore use the 2005 CICS to construct the core measure. For robustness, I also map sectors in the 2003 CICS to two-digit industries and compute intensity, finding a correlation of 0.97 with the 2005-based measure.⁵ In addition, I construct alternative measures from the 2002 and 2007 input–output tables by computing the public-sector consumption shares from other sectors.

Controls. Using the ASIF database, I compute pre-policy firm averages for size (log assets), log capital–labor ratio, age, and exporter status. Following [Liu et al. \(2023\)](#), I also control for major concurrent events to exclude their effects. First, to account for the reduction in trade policy uncertainty following the grant of PNTR to China by the United States Congress in October 2000, I include controls based on [Pierce and Schott \(2016\)](#), who show that such reductions had a positive impact on Chinese exports. Specifically, I calculate the normal-trade-relation (NTR) gap at the four-digit industry level and interact it with a post-PNTR dummy, which takes a value of one from 2001 onward (denoted $\text{TPU} \times \text{post01}$). Second, I construct interaction

⁵2003 CICS sectors—Garments, Food, Metals & machinery, Electronics, Chemicals & pharmaceuticals, and Auto & components—are mapped to two-digit industries 18&19, 14, 34, 40, 27, and 37, respectively.

terms between output tariffs and a post-2002 dummy (output tariff $\times$ *post02*), and between input tariffs and a post-2002 dummy (input tariff $\times$ *post02*), to control for average export tariff rates at the four-digit industry level, reflecting tariff reductions under most-favored-nation treatment.

5 Empirical Results

5.1 Baseline Results

Tables 1–3 report the baseline estimates of the effects of the GPL on different patent outcomes. Table 1 examines whether a firm applied for a utility model patent in year t (denoted *d_Utility*), while Table 2 examines the log count of utility model applications (denoted *ln Utility*). Column (1) includes only the core interaction term $gs \times post03$, firm fixed effects, and year fixed effects. Columns (2) and (3) add city $\times$ year and ownership $\times$ year fixed effects, respectively. Column (4) adds firm-level controls, and Column (5) adds controls for major concurrent events.

For the utility model dummy (Table 1), the coefficient on $gs \times post03$ is positive at the 10 percent level in Column (1) and remains positive and statistically significant at the 5 percent level as additional controls and fixed effects are introduced. By contrast, the coefficient in Table 2 is not statistically significant. This pattern indicates that the GPL’s effect is primarily on the extensive margin by encouraging more firms to apply for utility model patents, rather than increasing the number of applications per firm.

Table 3 extends the baseline analysis to invention and design patents. Columns (1) and (2) report results for an invention-patent dummy and the log count of invention patent applications, respectively; Columns (3) and (4) focus on design-patent outcomes. Across all specifications, the coefficient on $gs \times post03$ is statistically insignificant, suggesting that the GPL increased firms’ innovation levels mainly through applications for utility model patents, a patent type focusing on more functional improvement without substantial novelty or originality.⁶ All analysis afterward focuses on utility model patents.

⁶By definition, invention patents protect new technical solutions for products or processes with substantial novelty and inventiveness; utility model patents protect practical innovations focusing on functional improvements; and design patents protect the unique visual appearance or aesthetic design of a product.

Table 1. Baseline: Utility Model Patents (Dummy)

	(1)	(2)	(3)	(4)	(5)
$gs \times post03$	0.010* (0.006)	0.010* (0.005)	0.010** (0.005)	0.010** (0.005)	0.010** (0.005)
output tariff $\times post02$				0.000 (0.000)	0.000 (0.000)
input tariff $\times post02$				-0.001** (0.000)	-0.001** (0.000)
TPU $\times post01$				0.017 (0.014)	0.017 (0.014)
Controls	No	No	No	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
City-Year FE	No	Yes	Yes	Yes	Yes
Ownership-Year FE	No	No	Yes	Yes	Yes
Observations	1,022,580	1,022,580	1,022,580	1,022,580	1,022,580
Adj. R-squared	0.309	0.313	0.319	0.319	0.319

Notes: The dependent variable is a dummy equal to one if firm i files at least one utility model patent application in year t . The main explanatory variable, $gs \times post03$, is the interaction between industry-level procurement intensity gs_j and a post-2003 dummy. Columns (4)–(5) additionally include interactions of the industry-year output tariff and input tariff with a post-2002 dummy, and an interaction between trade policy uncertainty (TPU) and a post-2001 dummy. All regressions are estimated on 1,022,580 firm–year observations. Firm and year fixed effects are included in all columns; city–year fixed effects are added in columns (2)–(5), and ownership–year fixed effects in columns (3)–(5). Robust standard errors, clustered at the four-digit industry level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 2. Baseline: Utility Model Patents (Log Count)

	(1)	(2)	(3)	(4)	(5)
$gs \times post03$	0.020 (0.044)	0.036 (0.041)	0.062 (0.038)	0.062 (0.038)	0.060 (0.037)
output tariff $\times post02$				0.007* (0.004)	0.007* (0.004)
input tariff $\times post02$				-0.007 (0.006)	-0.007 (0.006)
TPU $\times post01$				-0.277 (0.230)	-0.277 (0.230)
Controls	No	No	No	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
City-Year FE	No	Yes	Yes	Yes	Yes
Ownership-Year FE	No	No	Yes	Yes	Yes
Observations	39,327	39,327	39,327	39,327	39,327
Adj. R-squared	0.465	0.465	0.472	0.472	0.472

Notes: The dependent variable is the log count of utility model patent applications filed by firm i in year t ($\ln_Utility$). The main explanatory variable, $gs \times post03$, is the interaction between industry-level procurement intensity gs_j and a post-2003 dummy. Columns (4)–(5) additionally include interactions of the industry-year output tariff and input tariff with a post-2002 dummy, and an interaction between trade policy uncertainty (TPU) and a post-2001 dummy. All regressions are estimated on 39,327 firm-year observations. Firm and year fixed effects are included in all columns; city-year fixed effects are added in columns (2)–(5), and ownership-year fixed effects in columns (3)–(5). Observations with zero utility model patent applications are omitted in this table. Robust standard errors, clustered at the four-digit industry level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 3. Baseline: Invention and Design Patents

	Invention		Design	
	$d_Invention$ (1)	$\ln Invention$ (2)	d_Design (3)	$\ln Design$ (4)
$gs \times post03$	0.003 (0.004)	-0.090 (0.068)	0.002 (0.001)	0.008 (0.058)
output tariff $\times post02$	-0.001*** (0.000)	0.005 (0.007)	0.000 (0.000)	0.004 (0.005)
input tariff $\times post02$	0.001 (0.000)	-0.016 (0.012)	-0.000 (0.000)	-0.010 (0.009)
TPU $\times post01$	0.004 (0.011)	0.307 (0.495)	0.012** (0.005)	0.044 (0.268)
Firm FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Ownership-Year FE	Yes	Yes	Yes	Yes
Observations	1,022,580	17,609	1,022,580	18,309
Adj. R-squared	0.255	0.525	0.269	0.503

Notes: The dependent variable is (i) a dummy equal to one if firm i files at least one invention patent application in year t in column (1), (ii) the log count of invention patent applications in year t in column (2), (iii) a dummy equal to one if firm i files at least one design patent application in year t in column (3), and (iv) the log count of design patent applications in year t in column (4). The main explanatory variable, $gs \times post03$, is the interaction between industry-level procurement intensity gs_j and a post-2003 dummy. All specifications include interactions of the industry-year output tariff and input tariff with a post-2002 dummy, and an interaction between trade policy uncertainty (TPU) and a post-2001 dummy. Firm fixed effects, city-year fixed effects, and ownership-year fixed effects are included in all columns. The number of observations differs across columns (as reported in the table). Robust standard errors, clustered at the four-digit industry level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

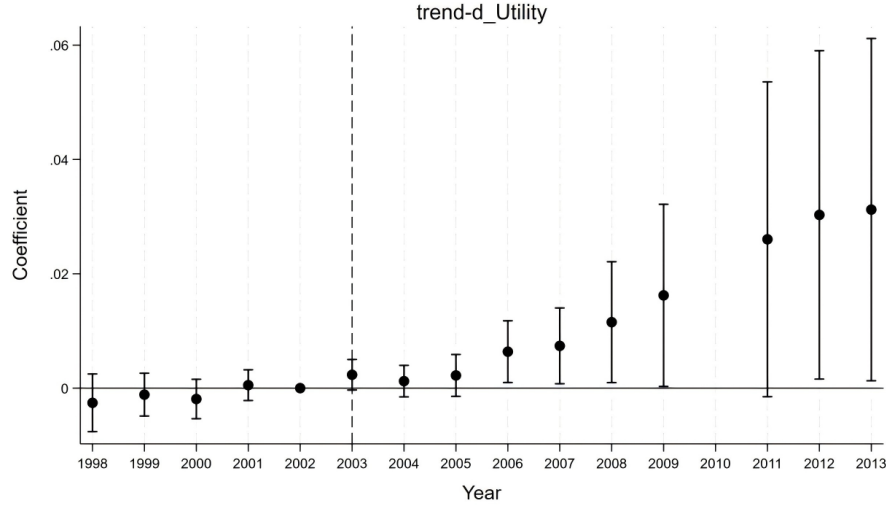


Figure 2. Event Study of the Dynamic Effects of the GPL

Notes: This figure shows the dynamic effects of the GPL. The x -axis denotes the year; the y -axis denotes the estimated coefficients $\hat{\beta}_k$ from equation (2). The benchmark year is 2002. The solid lines describe the 95% confidence intervals. Data for 2010 are omitted due to data problems.

I test the parallel trends assumption following equation (2) and present the estimates with 95 percent confidence intervals in Figure 2. The estimates are close to zero (relative to the benchmark year 2002) prior to 2003, which supports the identification strategy.

5.2 Robustness Checks

To address endogeneity concerns, I conduct several robustness checks. First, I check whether the results depend on the specific measure of industry procurement intensity. In the main analysis, I use the 2005 CICS data to construct the procurement intensity at the four-digit industry level, using a dummy gs to indicate whether an industry's procurement intensity is above the median. In Table 4, I replace this with the following alternatives: (i) the continuous value of procurement intensity (gs_value), (ii) procurement intensity at the two-digit industry level (gs_2digit), (iii) a procurement intensity dummy constructed from the 2002 input-output table (gs_IO02), and (iv) a dummy constructed from the 2007 input-output table (gs_IO07). In all cases, the interaction between procurement intensity and the post-2003 dummy remains positive and significant, indicating that the baseline findings are not sensitive to the specific measure of procurement intensity.

Table 4. Robustness Check: Alternative Measures of Procurement Intensity

	<i>gs_value</i> (1)	<i>gs_2digit</i> (2)	<i>gs_IO02</i> (3)	<i>gs_IO07</i> (4)
$gs \times post03$	0.473*** (0.160)	0.023*** (0.004)	0.014*** (0.004)	0.013*** (0.003)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Ownership-Year FE	Yes	Yes	Yes	Yes
Observations	1,022,580	1,022,580	1,022,580	1,022,580
Adj. R-squared	0.320	0.320	0.319	0.319

Notes: The dependent variable is a dummy equal to one if firm i files at least one utility model patent application in year t . The main explanatory variable is $gs \times post03$, the interaction between procurement intensity and a post-2003 dummy. Columns (1)–(4) use alternative measures of procurement intensity: continuous procurement intensity (*gs_value*), procurement intensity measured at the two-digit industry level (*gs_2digit*), a procurement intensity dummy constructed from the 2002 input–output table (*gs_IO02*), and from the 2007 input–output table (*gs_IO07*), respectively. All regressions include the baseline set of controls, firm fixed effects, city–year fixed effects, and ownership–year fixed effects. Robust standard errors, clustered at the four-digit industry level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Second, I perform additional robustness checks in Table 5: (1) trimming observations with procurement intensity in the top and bottom 1 percent; (2) excluding municipalities; (3) clustering standard errors at the firm level; and (4) clustering at the four-digit industry×city level. Across these specifications, the coefficient on $gs \times post03$ remains stable and significant.

Table 5. Other Robustness Checks

	1%–99% (1)	Excl. munis (2)	Clus. firm (3)	Clus. ind4-city (4)
$gs \times post03$	0.010** (0.005)	0.009* (0.005)	0.010** (0.004)	0.010** (0.005)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Ownership-Year FE	Yes	Yes	Yes	Yes
Cluster	Ind4	Ind4	Firm	Ind4–City
Observations	1,022,113	886,740	1,022,580	1,022,580
Adj. R-squared	0.319	0.315	0.319	0.319

Notes: The dependent variable is a dummy equal to one if firm i files at least one utility model patent application in year t ($d_Utility$). The main explanatory variable, $gs \times post03$, is the interaction between industry-level procurement intensity gs_j and a post-2003 dummy. Column (1) trims the sample to observations with procurement intensity between the 1st and 99th percentiles; column (2) excludes municipalities; columns (3) and (4) use alternative clustering schemes as indicated in the “Cluster” row. Time-varying firm-level controls, firm fixed effects, city-year fixed effects, and ownership-year fixed effects are included in all columns. Robust standard errors are reported in parentheses and are clustered as indicated. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.3 Heterogeneity Analysis

To explore heterogeneity in the GPL’s impact, I estimate triple-difference models that interact the core interaction term with pre-policy firm characteristics. Table 6 shows that the three-way interactions with indicators for prior R&D spending (d_RD), prior utility model applications ($d_Utility$), and above-median total factor productivity (d_TFP) are all significantly positive. These results indicate that the GPL has stronger effects on firms that were already investing in R&D, had past experience with utility model patents, or exhibited higher productivity before the GPL’s implementation. Since these firms are typically regarded as more efficient, this suggests that the GPL reduced contract misallocation, redirected resources to more efficient firms in a competitive procurement market, and thereby favored innovation.

Table 6. Heterogeneity: Prior R&D, Prior Utility Model Applications, and TFP

	<i>d_Utility</i>		
	(1)	(2)	(3)
$gs \times post03 \times d_{RD}$	0.017*** (0.006)		
$gs \times post03 \times d_{Utility}$		0.035*** (0.012)	
$gs \times post03 \times d_{TFP}$			0.013*** (0.003)
$gs \times post03$	0.005 (0.003)	0.010* (0.006)	0.003 (0.005)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes
Ownership-Year FE	Yes	Yes	Yes
Observations	1,219,844	1,220,154	1,213,117
Adj. R-squared	0.316	0.320	0.316

Notes: The dependent variable is a dummy equal to one if firm i files at least one utility model patent application in year t . The main explanatory variable, $gs \times post03$, is the interaction between industry-level procurement intensity gs_j and a post-2003 dummy. This table examines heterogeneity by interacting $gs \times post03$ with firm characteristics measured prior to the GPL: an indicator for whether the firm had any pre-GPL R&D expenditure (d_{RD}) in column (1), an indicator for whether the firm filed any pre-GPL utility model patent application ($d_{Utility}$) in column (2), and an indicator for whether the firm's average pre-GPL TFP is above the median (d_{TFP}) in column (3). All regressions include the baseline set of controls, firm fixed effects, city-year fixed effects, and ownership-year fixed effects. Robust standard errors, clustered at the four-digit industry level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

5.4 Mechanism Analysis

The GPL aims to standardize procurement procedures and foster fair competition, which should reduce rent-seeking, incentivize firms to invest in R&D inputs, and ultimately raise innovation. The effect could be more pronounced in previously corrupt regions. To test these mechanisms, in Table 7 I first replace the dependent variable with measures of R&D investment: the ratio of R&D to sales (k_{rd}) and the log of R&D spending ($\ln RD$) in Columns (1)–(2). In Columns (3)–(4), I estimate triple-difference specifications that interact the core term with city-level proxies for political connections: a dummy for cities with an above-median share of firms employing government-affairs specialists ($d_{specialstaff}$) and a dummy for cities with

above-median pre-policy management expenses (d_mfee). The results show that firms invest more in R&D after the GPL and that the impact is stronger in more politically connected regions, consistent with the proposed mechanisms.

However, direct evidence on contract allocation is unavailable due to data constraints. I therefore propose a research design to support more granular mechanism analysis, discussed in detail in Section 7.

Table 7. Mechanism Analysis: R&D Investment and Corruption

	k_rd (1)	$\ln RD$ (2)	$d_Utility$ (3)	$d_Utility$ (4)
$gs \times post03$	0.001* (0.000)	0.063* (0.038)	0.007 (0.005)	0.005** (0.002)
$gs \times post03 \times d_specialstaff$			0.006** (0.003)	
$gs \times post03 \times d_mfee$				0.008* (0.005)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes
Ownership-Year FE	Yes	Yes	Yes	Yes
Observations	742,045	130,502	811,178	1,220,154
Adj. R-squared	0.434	0.780	0.324	0.315

Notes: The dependent variable is the ratio of R&D expenditure to sales (k_rd) in column (1), the log of R&D spending ($\ln RD$) in column (2), and a dummy equal to one if firm i files at least one utility model patent application in year t in columns (3)–(4). The main explanatory variable, $gs \times post03$, is the interaction between industry-level procurement intensity gs_j and a post-2003 dummy. Column (3) estimates a triple-difference specification by additionally interacting $gs \times post03$ with $d_specialstaff$, a city-level dummy for cities with an above-median share of firms employing government-affairs specialists. Column (4) analogously interacts $gs \times post03$ with d_mfee , a city-level dummy for cities with above-median pre-policy management expenses. All regressions include the baseline set of controls. Firm fixed effects, city-year fixed effects, and ownership-year fixed effects are included in all columns. The number of observations differs across columns (as reported in the table). Robust standard errors, clustered at the four-digit industry level, are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

6 Conclusions

Using the 2003 Government Procurement Law as a quasi-natural experiment, I examine the effects of government procurement regulation on firms' innovation. I find a positive and persistent effect of the GPL on firms' innovation growth. After the im-

plementation of the GPL, firms’ innovation levels in procurement-intensive industries significantly increased. The effect is greater for firms with prior R&D expenditures, prior patent application experience, or higher total factor productivity, which tend to be more efficient firms. I further observe that firms’ R&D investment in procurement-intensive industries increased compared with their counterparts, and that the effect is stronger in more corrupt regions. This pattern is consistent with the objective of the GPL, which was to promote an open procurement market and reallocate government contracts toward more efficient firms, particularly in places that were previously more corrupt and distorted. By doing so, the GPL has strengthened firms’ incentives to invest in R&D, and fostered innovation growth.

7 Future Research Proposal

My future research continues the investigation of how corruption in contract allocation affects firm innovation, but in a more recent context. I have constructed a novel contract-level dataset that enables accurate identification of corrupt contracts and granular tracking of firm performance. The database is scraped from a provincial procurement website and contains rich information on bidding companies (including expert scores of firms’ products), covering approximately 50,000–80,000 observations per year from 2015 onward.

I will first detect corrupt contracts, following the method proposed by [Chen \(2025\)](#). This method is based on the observation that, while corruption was largely restrained in response to the GPL, it has reemerged in a more covert form in recent years: scoring rules can be manipulated to favor predetermined suppliers, and the minimum-number-of-bidders requirement is often satisfied with “zombie bids” (firms that accompany the predetermined winner to create a facade of competition but have virtually no chance of winning). If the winner’s actual score significantly exceeds the expected maximum scores of its strongest rivals, i.e., if the winner leaves an unusually large profit cushion, this is taken as a red flag for potential corruption in that auction. Using this criterion, [Chen \(2025\)](#) develops a statistical detection method that has a correct rate exceeding 94%, benchmarked against real specialist evaluations.

Using this method, I will analyze the following stylized facts: (i) How prevalent is scoring-rule-manipulation corruption in open auctions? (ii) Who is more likely to engage in corruption? Specifically, are corrupt firms more politically connected, and are small and medium-sized enterprises more disadvantaged in this process? (iii) How

is corruption correlated with technological innovation? In particular, do corrupt firms invest less in R&D?

This dataset can also be used to analyze the causal relationship between obtaining government procurement contracts and firms’ subsequent innovation. Theoretically, the impact of procurement contracts on firm innovation is ambiguous. On the one hand, government contracts could have positive signals for firm quality or future cash flow, thus attracting external funding, high-skill workers, and customers (Howell, 2017; Lerner, 1999; Goldman and Strahan, 2020; Gabriel, 2024); or provide government resources that are shareable for commercial uses, bringing about economies of scale (Beraja et al., 2023). On the other hand, however, government contracts always emphasize lower prices, squeezing the profit space of firms (Cao et al., 2024); government contracts also prioritize off-the-shelf goods, with limited requirement of innovation (Howell et al., 2021); in addition, government is found to selectively delay payment to small and medium-sized enterprises during China’s deleveraging, worsening their financial conditions and operating performance (Hu et al., forthcoming). These factors are in turn discouraging innovation. My proposed research uses a quasi-natural experiment of contract awarding in close auctions to better understand the causal effects and mechanisms on the topic.

I will identify the causal effects of obtaining procurement contracts on firms’ subsequent innovation in a quasi-exogenous context of “close auctions”, i.e. winners and runners-up have very close scores, and the contract award is therefore quasi-random. Specifically, I identify close auctions in all procurement contracts defined by the win margin⁷, and retain only winners and runners-up who have not previously won an award. The use of close auctions is critical for the empirical design, largely mitigating selection concerns. If one directly compares winners and losers, there could be severe bias, for example if politically connected firms are both more likely to win an award and weaker in innovation. I also focus on firms without previous awards to get a more homogeneous sample, as well as to isolate the effect of the first contract. I further conduct rich mechanism analysis to investigate several possible channels, such as signaling effects, capital provision, R&D investment, and political connections.

⁷For example, when

$$\text{score_margin}_k = \frac{\text{winner_score}_k - \text{second_score}_k}{\text{second_score}_k} < 0.5\%.$$

The specific bandwidth is guided by the empirical distribution of margins and is allowed to vary in robustness checks.

References

- Acemoglu, Daron, Ufuk Akcigit, Harun Alp, Nicholas Bloom, and William Kerr**, “Innovation, Reallocation, and Growth,” *American Economic Review*, 2018, 108 (11), 3450–3491.
- Baltrunaite, Audinga**, “Political Contributions and Public Procurement: Evidence from Lithuania,” *Journal of the European Economic Association*, 2020, 18 (2), 541–582.
- Beraja, Martin, David Y. Yang, and Noam Yuchtman**, “Data-Intensive Innovation and the State: Evidence from AI Firms in China,” *Review of Economic Studies*, 2023, 90 (4), 1701–1723.
- Bosio, Erica, Simeon Djankov, Edward Glaeser, and Andrei Shleifer**, “Public Procurement in Law and Practice,” *American Economic Review*, 2022, 112 (4), 1091–1117.
- Brugués, Fernando, Joaquim Brugués, and Stefano Giambra**, “Political Connections and Misallocation of Procurement Contracts: Evidence from Ecuador,” 2022. Working paper.
- Cao, Shengmao, Xuejie Yi, and Chuan Yu**, “Competitive Bidding in Drug Procurement: Evidence from China,” *American Economic Journal: Economic Policy*, 2024, 16 (3), 481–513.
- Chen, Qian Michelle**, “Rigging the Scores: Corruption through Scoring Rule Manipulation in Public Procurement Auctions,” 2025. Unpublished working paper.
- Chen, Zhihong and Jing Zhang**, “Types of Patents and Driving Forces Behind the Patent Growth in China,” *Economic Modelling*, 2019, 80, 294–302.
- Chen, Zhiyong, Zhi Liu, Juan Carlos Suárez Serrato, and Daniel Yi Xu**, “Notching R&D Investment with Corporate Income Tax Cuts in China,” *American Economic Review*, 2021, 111 (7), 2065–2100.
- Coviello, Decio and Stefano Gagliarducci**, “Tenure in Office and Public Procurement,” *American Economic Journal: Economic Policy*, 2017, 9 (3), 59–105.
- Ericson, Richard and Ariel Pakes**, “Markov-Perfect Industry Dynamics: A Framework for Empirical Work,” *Review of Economic Studies*, 1995, 62 (1), 53–82.
- Fang, Lily Hua, Josh Lerner, and Chaopeng Wu**, “Intellectual Property Rights Protection, Ownership, and Innovation: Evidence from China,” *Review of Financial Studies*, 2017, 30 (7), 2446–2477.
- Gabriel, Ricardo Duque**, “The Credit Channel of Public Procurement,” *Journal of Monetary Economics*, 2024, 147 (S), 103601.

- Goldman, Jim and Philip E. Strahan**, “Government as Customer of Last Resort: The Stabilizing Effects of Government Purchases on Firms,” *Review of Financial Studies*, 2020, 33 (2), 610–643.
- Hang, Jie and Chao Zhan**, “Government Procurement and Resource Misallocation: Evidence from China,” *Journal of Economic Behavior & Organization*, 2023, 216, 568–589.
- Hopenhayn, Hugo A.**, “Entry, Exit, and Firm Dynamics in Long Run Equilibrium,” *Econometrica*, 1992, 60 (5), 1127–1150.
- Hopenhayn, Hugo and Richard Rogerson**, “Job Turnover and Policy Evaluation: A General Equilibrium Analysis,” *Journal of Political Economy*, 1993, 101 (5), 915–938.
- Howell, Sabrina T.**, “Financing Innovation: Evidence from R&D Grants,” *American Economic Review*, 2017, 107 (4), 1136–1164.
- , **Jason Rathje, John Van Reenen, and Jun Wong**, “Opening Up Military Innovation: Causal Effects of ‘Bottom-Up’ Reforms to U.S. Defense Research,” NBER Working Paper 28700, National Bureau of Economic Research April 2021. Revised September 2025.
- Hu, Jiayin, Songrui Liu, Yang Yao, and Zhu Zong**, “Government Deleveraging and the Reverse Crowding-in Effect: Evidence from Subnational Debt and Government Contractors,” *Management Science*, forthcoming.
- Jovanovic, Boyan**, “Selection and the Evolution of Industry,” *Econometrica*, 1982, 50 (3), 649–670.
- König, Michael, Kjetil Storesletten, Zheng Song, and Fabrizio Zilibotti**, “From Imitation to Innovation: Where Is All That Chinese R&D Going?,” *Econometrica*, 2022, 90 (4), 1615–1654.
- Kou, Zonglai and Xueyue Liu**, “Chinese Firms’ Patent Behavior: Stylized Facts and Policy Impacts,” *Economic Research Journal (Jingji Yanjiu)*, 2020, 55 (3), 83–99. In Chinese.
- Lerner, Josh**, “The Government as Venture Capitalist: The Long-Run Impact of the SBIR Program,” *Journal of Business*, 1999, 72 (3), 285–318.
- Li, Xuan**, “Behind the Recent Surge of Chinese Patenting: An Institutional View,” *Research Policy*, 2012, 41 (1), 236–249.
- Li, Zeren and Songrui Liu**, “Favor with Fear: Risk-driven Favoritism in Government Procurement in China,” 2024. Working paper.

- Liu, Xueyue, Yu Liu, and Ruili Zhao**, “Import Competition and Energy Efficiency: Firms’ Responses to the WTO Accession in China,” *Journal of Economic Behavior & Organization*, 2023, *214*, 670–690.
- OECD**, “Preventing Corruption in Public Procurement,” Technical Report, OECD Publishing, Paris 2016.
- Pierce, Justin R. and Peter K. Schott**, “The Surprisingly Swift Decline of U.S. Manufacturing Employment,” *American Economic Review*, 2016, *106* (7), 1632–1662.
- Shao, Yichen and Chun Xiao**, “Corporate Tax Policy and Heterogeneous Firm Innovation: Evidence from a Developing Country,” *Journal of Comparative Economics*, 2019, *47* (2), 470–486.
- Titl, Vaclav and Benny Geys**, “Political Donations and the Allocation of Public Procurement Contracts,” *European Economic Review*, 2018, *111*, 443–458.
- Wu, Chaopeng and Di Tang**, “IPR Enforcement, Technological Innovation, and Firm Performance: Evidence from Chinese Listed Firms,” *Economic Research Journal (Jingji Yanjiu)*, 2016, *51* (11), 125–139. In Chinese.